# LEARNING OUTCOMES!

**Candidates should be able to:**

- identify on a map the Mangla, Tarbela and Warsak dams, and name two examples of barrages
- understand the importance of water as a resource; understand how supplies for agricultural, industrial and domestic purposes are obtained, maintained and controlled as well as used; understand the reasons for, and consequences of, the Indus Water Treaty
- explain and evaluate the causes of and solutions to the problems of water supply (including pollution)
- understand the value of water as a resource for development
- explain and evaluate how water supply issues can lead to conflict.
- describe the different types of irrigation and explain the advantages and disadvantages of each for small-scale subsistence farming, and for the growing of cotton, rice, sugar cane and wheat: – canal irrigation – karez, inundation and perennial canal – lift irrigation – Persian wheel and tubewell
- understand the roles of dams, barrages, link canals, distribution canals, field channels and bunds
- explain the causes of waterlogging and salinity, and:
- explain how land damaged by it can be restored
- evaluate how agricultural practice and water management can be improved to prevent it happening

# IMPORTANCE OF RIVER

- ✓ Adds to scenic beauty
- ✓ Provides water to even those areas where rainfall is extremely low
- ✓ Helps to generate hydroelectricity
- ✓ Increases fertility of land by carrying alluvium and organic matter
- ✓ Fishing is practical in rivers
- ✓ Provides water for domestic and industrial purposes
- ✓ Supplies water for irrigation

# USES OF WATER

## DOMESTIC USES OF WATER

- ✓ Drinking
- ✓ Bathing
- ✓ Watering the yard and gardens
- ✓ Preparing food
- ✓ Washing clothes and dishes

## INDUSTRIAL USES OF WATER

- ✓ Pharmaceutical Industries for injections, syrups
- ✓ Tanning industry for washing, dyeing
- ✓ Food processing industry for preparing juices and beverages
- ✓ Textile Industry for washing, bleaching, blurring, dyeing
- ✓ Iron and Steel industry to cool down furnace for making steel
- ✓ Thermal power station to produce steam that makes the turbines move

## AGRICULTURAL USES OF WATER

- ✓ For irrigation
- ✓ For Buffaloes
- ✓ For Food of Livestock

# CONFLICTS OVER WATER

- ✓ Domestic vs Industry vs Agriculture
- ✓ Punjab vs Sindh
- ✓ India vs Pakistan (1947-1960)

# INDUS WATER TREATY

- ✓ Most of Pakistan suffers from low rainfall and unreliable rainfall
- ✓ Increasing population means more food is needed
- ✓ Could only be provided by irrigated land
- ✓ India cut off water supplies to Pakistan so famine starvation, crops failed and land became arid
- ✓ Threatened Pakistan
- ✓ Pakistan made to buy water from India
- ✓ Treaty included construction of Tarbela and Mangla dams, construction of 5 barrages, remodeling of existing canals and head works and construction of eight link canals

## Importance

- Ensures that India does not restrict Pakistan's water supply. Water supply in Pakistan is maintained
- Pakistan has now exclusive rights to waters of the rivers Indus, Jhelum, and Chenab
- Maintains agricultural production
- Tarbela and Mangla dams built [to store water]
- Barrages / link canals built [to distribute water]
- Enabled construction cost of works to be shared with Western countries and India

# IRRIGATION

- ✓ Artificial supply of water

# MODERN METHODS

### Advantages

- ✓ More efficient / faster / does not need to rest
- ✓ For larger area / more water / goes deeper
- ✓ Regular supply / can be used at any time of year / continuous
- ✓ Less labour required
- ✓ Cleaner water
- ✓ Reduces waterlogging and salinity (only tubewells)

### Disadvantages

- ✓ Expensive / cannot be used by poor farmers
- ✓ High Maintenance cost
- ✓ Needs fuel /electricity / diesel etc.
- ✓ Reduces groundwater / lowers water table

# TRADITIONAL METHODS

### Advantages

- ✓ Cheap / can be used by poor farmers
- ✓ Low Maintenance cost
- ✓ Does not need fuel /electricity / diesel etc.
- ✓ Do not reduce groundwater / lowers water table

### Disadvantages

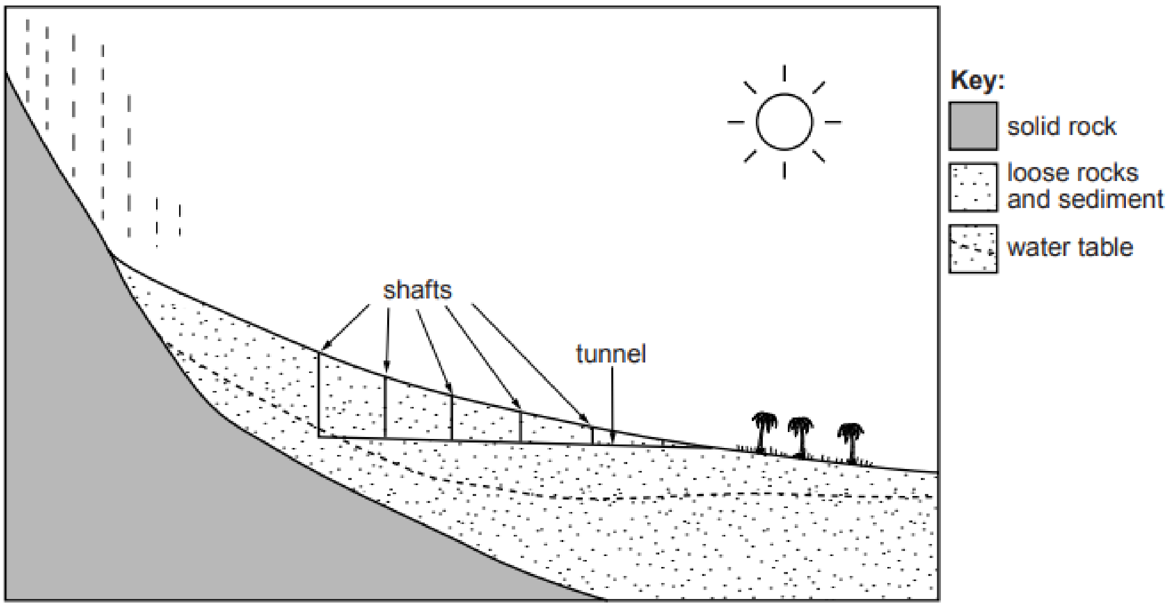
- ✓ Less efficient / slower
- ✓ For smaller area / less water
- ✓ No Regular supply / cannot be used at any time of year
- ✓ More labour required
- ✓ Unclean water
- ✓ Waterlogging and salinity

# WHY IRRIGATION IS NEEDED?

- ✓ Insufficient annual rainfall
- ✓ Most of Pakistan is arid
- ✓ Monsoon is variable in amount, timings and distributions
- ✓ High temperature condition which leads to a great degree of evapotranspiration
- ✓ 1/3rd of country in south has less than 10 rainy days in a year
- ✓ 2/3rd has less than 20 rainy days in a year

# KAREZ

- ✓ When rainfall occurs in mountains, water sinks into the ground
- ✓ Karez is a water tunnel or narrow underground canal
- ✓ It starts from the base of hill or mountain or run 1-2 km underground before it emerges above ground
- ✓ Length depends on the distance between source of Karez and the command area
- ✓ Throughout its length it is dotted with vertical shafts which are used to clean and repair it
- ✓ Selecting of site for digging is done by experienced village elders
- ✓ Digging and repair is done by a group of trained labors
- ✓ It is privately owned by group of people rather than a single person
- ✓ Owner's share water according to percentage share in Karez
- ✓ Karez are drying out in areas where there are no tube wells
- ✓ Karez are drying because they are neglected



# PERSIAN WHEEL

- ✓ Powered by blind-folded bullock using wooden shaft
- ✓ It turns a horizontal wooden wheel
- ✓ It is geared to a vertical wheel at the end of the shaft
- ✓ This carries the vertical metal wheel which is attached to chain of pots
- ✓ Pots raise water from the well and spill their contents in to channel that leads to the fields

# SHADUF

- ✓ Consists of bucket suspended by a rope from one end of a pole
- ✓ Weight is placed at the other end of the pole
- ✓ The pole is suspended on a y-shaped post at a well or a river bank
- ✓ The bucket is dipped in to water by hand and weight at the other end of the pole helps to lift it up
- ✓ One tenth of a hectare can be irrigated daily

# TANK IRRIGATION

- ✓ Tank is a reservoir of any specific size
- ✓ There are practised by constructing mud banks across small streams or constructed across slopes for collecting and preserving rain water and surface runoff from mountain slopes
- ✓ Preserve water for dry season
- ✓ Occupy large area
- ✓ Evaporation

# INUNDATION CANALS

- ✓ Long canals taken off from large rivers
- ✓ They receive water when river is high enough and especially when it is flood
- ✓ Active in summers
- ✓ No control over water supply
- ✓ Irrigates small area
- ✓ Taken out from IJCRS
- ✓ No waterlogging and salinity because evaporation rate is higher

# DIVERSION CANALS

- ✓ Narrow version of an inundation canal
- ✓ Water is taken off from narrow streams through small man-made channels often high up on valley sides to small terraced fields
- ✓ Irrigates small area
- ✓ Practiced only when river is full
- ✓ Practiced in northern mountains

# PERENNIAL CANALS

- ✓ Canals which supply water to their commercial areas throughout the year
- ✓ Supply water throughout the year
- ✓ Water is always available when needed
- ✓ Water can be better controlled
- ✓ Reliable
- ✓ Fills rainfall gaps
- ✓ Irrigates large areas
- ✓ Non fuel costs
- ✓ Cheap to use
- ✓ Evaporation losses
- ✓ Seepage losses if canals are unlined
- ✓ Cost of construction is high
- ✓ Deposition of silt
- ✓ Growth of dengue
- ✓ Need extra costs to treat algae growth

## Explain how perennial canals can damage farmland?

- ✓ Too much water

- ✓ Water table Rises
- ✓ Water logging
- ✓ Water Evaporates
- ✓ Causes Salinity

### Why perennial canals better than inundation canals?

- ✓ Water is always available when needed
- ✓ Water can be better controlled
- ✓ Reliable

# LINK CANALS

- ✓ Take water from western river for eastern river
- ✓ To compensate for water lost to India from eastern river
- ✓ Chasma – Jhelum (Indus to Jhelum)
- ✓ Rasul – Qadirabad (Jhelum to Chenab)
- ✓ Qadirabad – Balloki (Chenab to Ravi)
- ✓ Balloki – Sulamanki (Ravi to Sutlej)

# TUBEWELLS

- ✓ Long metal tube is drilled into the ground till it reaches the underground aquifer
- ✓ Water is then pumped up
- ✓ In drier regions that don't receive sufficient rainfall and lack irrigation systems, tube wells are often attached to motor pumps, which are operated by electricity or diesel.
- ✓ When water is pumped up it flows in to ponds from where it is distributed to fields by canals pipes or sprinklers
- ✓ They can tap water from depth of 92m or more to irrigate farms of more than 1000 nectars.
- ✓ Punjab is leading user of tube wells
- ✓ Irrigates large area
- ✓ No labour
- ✓ Continuous supply
- ✓ Reduces water logging and salinity
- ✓ Control of water
- ✓ Water available throughout the year
- ✓ Increases yield
- ✓ Lower water table
- ✓ Hugh cost of installation High maintenance cost
- ✓ Lack of electricity
- ✓ Lack of technology for pumps
- ✓ Deplete ground water
- ✓ Limited to large scale commercial farms

# SPRINKLER / SPRAY IRRIGATION

- ✓ Pressurized water from public pipes is sprayed over plants to provide then with water
- ✓ This system can be of any size, ranging from a home sprinkler used to keep a lawn green to industrial sized sprinklers used to irrigate crops
- ✓ Some sprinklers heads only spray in one direction while others rotate as they spray
- ✓ Very less water loss
- ✓ Can be used according to farmer's requirement can be well controlled
- ✓ No labour required
- ✓ Equal distribution of water
- ✓ More efficient
- ✓ Less time consuming
- ✓ Expensive to install
- ✓ Requires pumping, electricity is expensive
- ✓ Irrigates limited land
- ✓ Lack of technology
- ✓ High winds can drift spray

# TANKERS

- ✓ Tankers collect water from ponds, lakes and ground water and provide it to households, fields and linear plantations
- ✓ Expensive


WATER

# RIVERS

- ✓ Indus
- ✓ Jhelum
- ✓ Chenab
- ✓ Ravi
- ✓ Sutlej
- ✓ Kabul

# DAMS

- ✓ Tarbela (Indus)
- ✓ Mangla (Jhelum)
- ✓ Warsak (Kabul)

| SMALL DAMS | LARGE DAMS |
|---|---|
| Store water for irrigation | Store water for irrigation |
| Irrigates local areas only | Irrigates a vast area |
| Supply water for industrial and domestic use | Supply water from industrial and domestic use |
| Supply little or no electricity | Major suppliers of HEP |
| Silting problem easier to solve | Silting problem difficult to solve |
| Low initial investment | High initial investment |
| Maintenance cost is low | Maintenance cost is high |
| Less construction time | More construction time |
| Less important for flood | More important for flood |

### Advantages of Building Dams

- ✓ HEP
- ✓ Electrification
- ✓ Will extend life of Tarbela and Magla Dam (low pressure on them )
- ✓ Prevents flooding
- ✓ Development in region
- ✓ Possibility of Tourism
- ✓ Source of employment
- ✓ Water for irrigation

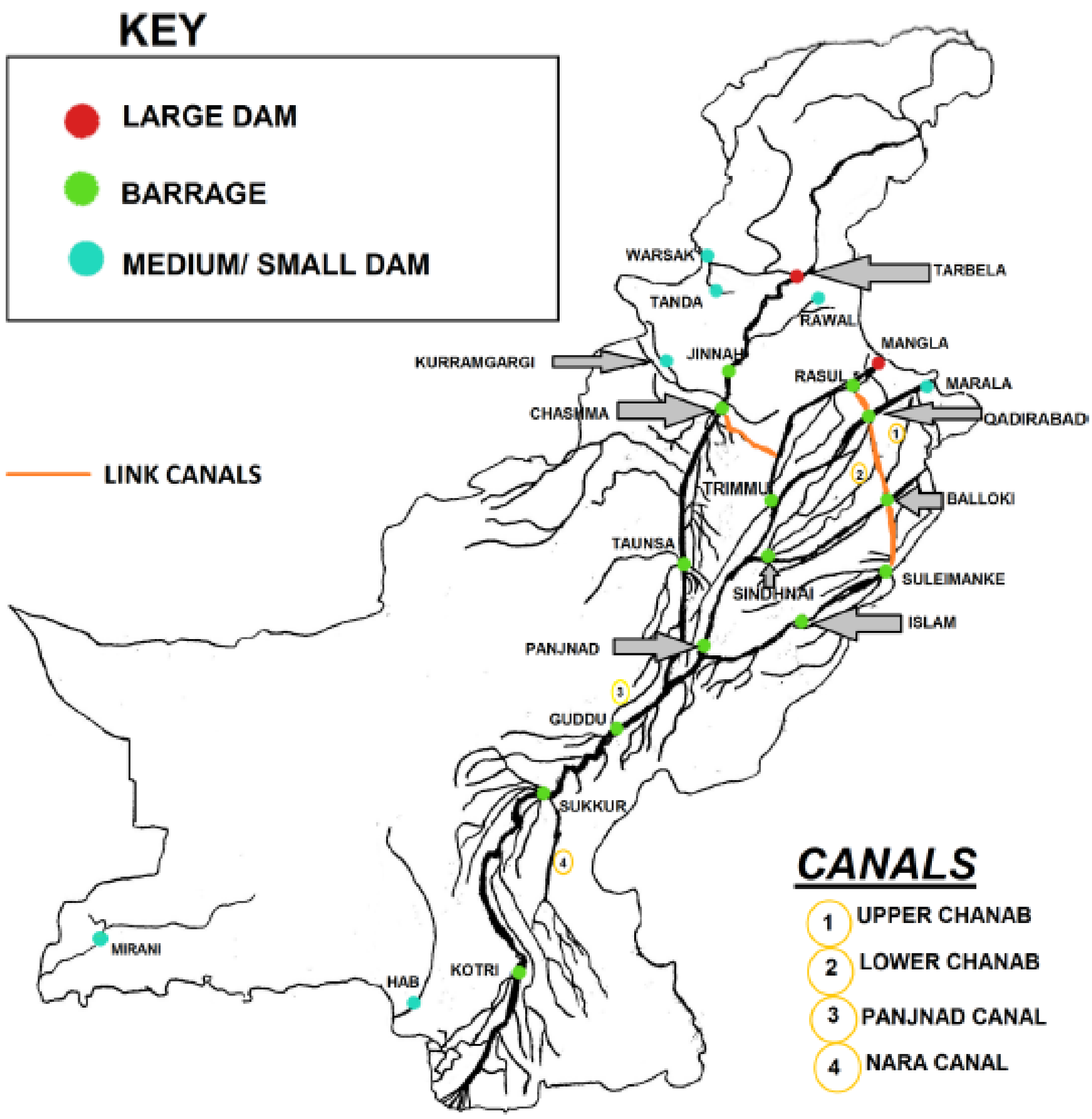
# BARRAGES

- ✓ Large structure used for irrigation and flood control
- ✓ Not involved in the generation of electricity
- ✓ Cost of construction is less than that of dam
- ✓ Can be made even in flat areas
- ✓ Size and capacity of barrage depends on the width of the river
- ✓ Sukkur Barrage on River Indus
- ✓ Guddu Barrage on River Indus
- ✓ Kotri Barrage on River Indus
- ✓ Rasul Barrage on River Jhelum

- ✓ Marala Barrage on River Chenab
- ✓ Balloki Barrage on River Ravi
- ✓ Sulaimanki Barrage on River Sutlej

## Comparision of Barrages with Dams

- ✓ Dams store water, Barrages control flow of water
- ✓ Dams produce HEP, Barrages do not.
- ✓ Dams face silting problem, barrages do not.
- ✓ Dams construction cost is higher than that of barrages.
- ✓ Dams require steep slopes to be built, Barrages can be built on flat lands.
- ✓ Dam is less wider than Barrage
- ✓ Dam is higher than barrage in height.



# WATERLOGGING AND SALINITY

- ✓ An irrigated area is said to be waterlogged when subsoil water rises high
- ✓ Caused by too much irrigation water
- ✓ Water table rises
- ✓ Soil becomes too wet, forming, puddles of water
- ✓ Water rises up wards carrying salts
- ✓ Evaporation causes salinity
- ✓ Salt patches
- ✓ Salt poisons crops so they die


## Why Problem For Farmers?

- ✓ Reduces cultivable area
- ✓ Reduces yield
- ✓ Reduces income & profit
- ✓ Expensive to reclaim land

**When asked for 6 marks, also mention how waterlogging and salinity occurs.**

## Solution

- ✓ Use of gypsum
- ✓ Canal closures to control volume of water

- ✓ Planting eucalyptus tree to lower water underground
- ✓ Surface drain
- ✓ Planned closures
- ✓ Lining of canals
- ✓ Installing tube wells
- ✓ Education/Training
- ✓ Govt schemes SCARP,WAPDA

# WATER POLLUTION

## Causes

- ✓ Sewage discharged into rivers
- ✓ Domestic waste thrown into rivers
- ✓ Pesticides/fertillisers runoff from agricultural fields
- ✓ Industrial waste thrown into rivers
- ✓ No organized collection of domestic waste
- ✓ Waste from ship discharged into rivers
- ✓ Leakage of oil from ship

## Effects

- ✓ Not for drinking / poisonous / contaminates groundwater
- ✓ Cost of treatment
- ✓ Causes disease – risk of cholera, typhoid, diarrhoea , hepatitis, dysentery etc.
- ✓ Not for food processing (e.g. fish canning)
- ✓ Smells
- ✓ Reduces fish catch / kills fish
- ✓ Can damage machinery
- ✓ Blocks canals / causes flooding
- ✓ Risk of malaria from stagnant water

## Solutions

- ✓ Treatment of sewage
- ✓ Improve sanitation facilities in poor quality housing
- ✓ Proper dumping of domestic and industrial waste
- ✓ Organized collection of Domestic waste
- ✓ Organic farming (Alternative to chemical fertillisers/pesticides)
- ✓ Fines on Water polluters
- ✓ Maintenance of ships

# SILTATION

- ✓ Deposition of rock/sand/silt in river/dams/barrages

## Causes

- ✓ Deforestation
- ✓ Destruction of mountains by agents of erosion
- ✓ Unlined canals

## Effects

- ✓ Blockage of canals
- ✓ Weakens the foundation of dams
- ✓ Croaking of irrigation canals
- ✓ Reduces capacity of reservoir
- ✓ Less HEP
- ✓ Less water for irrigation

## Solutions

- ✓ Afforestation
- ✓ Reforestation
- ✓ Lined canals
- ✓ Installation of silt traps
- ✓ Raise height of dam
- ✓ Dredging (Removal of Silt)
- ✓ Reducing wastage/pollution

# HOW WATER IS LOST?

- ✓ Seepage from beds of canals/absorbed into the soil/land/no canal lining;
- ✓ Evaporation/evapotranspiration from surface of canals/tanks/flooded land;
- ✓ Excessive runoff of water immediately into streams/rivers;
- ✓ Theft of water/theft from canals;
- ✓ Water drawn up by vegetation on side of canal;
- ✓ Mismanagement.

# WHY WATER DECREASING IN RESERVOIR?

- ✓ Siltation
- ✓ Soil Erosion
- ✓ Lower Rainfall
- ✓ Higher Temperature
- ✓ More Evaporation
- ✓ More usage

## Solution

- ✓ Afforestation
- ✓ Reforestation
- ✓ Lined canals
- ✓ Installation of silt traps
- ✓ Raise height of dam
- ✓ Dredging (Removal of Silt)
- ✓ Reducing wastage/pollution

# HOW TO INCREASE WATER SUPPLY?

- ✓ Rivers available
- ✓ Rainfall in monsoon
- ✓ Glaciers melt so water from mountains
- ✓ Use Flat land for canals
- ✓ Small Dams should be developed to store surplus flow during monsoon season
- ✓ Canals should be lined
- ✓ Fresh – water resources should not be used as dumping sites of solid and liquid waste
- ✓ Controlling seepage of toxic waste in to ground
- ✓ Desalination of sea water
- ✓ Awareness

# PROBLEMS IN INCREASING WATER SUPPLY?

- ✓ Not enough river water
- ✓ Not enough rain
- ✓ Loss by leakage, siltation
- ✓ Indus Water Treaty restricts water in reservoirs/rivers
- ✓ Evaporation in hot climate
- ✓ Water pollution
- ✓ Demands always increasing
- ✓ some places remote (e.g. Baluchistan)
- ✓ Lack of funds/government will
- ✓ Cost of reservoirs, canals etc
- ✓ Cost of tubewells
- ✓ Lack of reservoirs / dams / barrages
- ✓ Waterlogging and salinity problems

# WHY DEMANDS OF USERS CAN NOT BE MEET?

- ✓ Shortage of rainfall
- ✓ Evaporation
- ✓ More dams on rivers
- ✓ Canal system does not reach all those who need it
- ✓ Siltation in reservoirs
- ✓ Seepage / leakage from canals

- ✓ Wastage by users. Some use more than they need
- ✓ Water pollution
- ✓ High demand
- ✓ Theft of water
- ✓ Population increase
- ✓ Lack of investment

# 6 MARKS QUESTIONS

**Question 1** **REVIEW BOOK BY MYM**

**Read the following two views about water shortages in Pakistan:**

**A**

The best way to prevent water shortages in Pakistan is to build more dams and other infrastructure projects. These will store or supply more water.

**B**

The best way to prevent water shortages in Pakistan is to educate people about different methods of saving water. These methods could be carried out in agriculture, industry, and in homes.

***Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider both View A and View B in your answer. [6]***

I agree with view B more.

Why?

- ✓ High initial investment needed to build dams whereas to save water we can take following steps which are all cheap
  - Planting trees
  - Lining canals
  - Careful monitoring/regulation of amount of water used
  - Better forms of water storage in homes
  - Water meters in homes/industries
- ✓ Conflict between countries and provinces occur when constructing new Dams. For Example Kalabagh Dam.
- ✓ Building of dams take a lot of time as Mangla and Tarbela were proposed to be built in 1960 but they were finalized after almost 15 years
- ✓ Loss of fresh water at Indus Delta due to building of dams so Fishing industry suffer
- ✓ Siltation occurs in Dam due to our neglections so it do not benefit us a lot.

**Question 2** **REVIEW BOOK BY MYM**

**Read the following two views about providing solutions to the challenges of water supply in Pakistan:**

A

Small dams are the best way to solve water shortage problems and assist development.

B

Large dams are the best way to solve water shortage problems and assist development.

**Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider View A and View B in your answer. [6]**

I agree with view A more.

Why?

- ✓ Small dams need low initial investment whereas large dams are too expensive to be built.
- ✓ Less technical expertise needed in small dams which creates less burden on economy.
- ✓ Large dams require too much time to be built whereas small dams can be built too quickly.
- ✓ For large dams bigger area is needed so people and environment both are damaged to very large extent whereas small dams create minimal disruption to people and environment.
- ✓ Large dams are expensive so it can be built at few places which creates social issues as water is unevenly distributed so in my opinion small dams can reduce these conflicts.
- ✓ Large dams produce more HEP. However there are chances of power theft or power loss during transmission as long transmission lines are setup.